

Climate 5000 Series - Gen 4

18K BTU Single Zone Regular System

Wall Mounted IDU

Engineering
Submittal
Sheet



BOSCH

Overview and Project Information



Project name	
Location	
Architect	
Engineer	
General contractor	
Reference #	

18K Btu Single Zone Regular System - Wall Mounted IDU

Indoor part number		8-733-962-674
Indoor model number		BMS500-AAU018-1AHWXD
Outdoor part number		8-733962-686
Outdoor model number		BMS500-AAS018-1CSXRD
Efficiency Data		AHRI 210/240 2023
SEER2		23.5
EER2		12.7
HSPF2 Region IV		10.8
HSPF2 Region V		8.0
COP(47F)		3.50
COP (5F)		2.10
Performance Data		AHRI 210/240 2023
Cooling	Capacity Range	3500~18900
	Rated Capacity	18000
	Current	6.3
	Input Power	1406
Heating	Capacity Range	7700~21000
	Rated Capacity (47F)	18000
	Rated Capacity (5F)	12800
	Current	6.7
	Input Power	1507
Electrical Data		
Power supply (V / Ø / Hz)		208-230 / 1 / 60
Min. circuit ampacity indoor/outdoor (A)		3.0 / 18.0
Max. fuse indoor/outdoor (A)		15.0 / 20.0
Compressor Data		
Type:		Rotary
Refrigerant oil /oil charge (ml)		VG74 / 440
Input (W)		1090
Rated current (RLA/A)		7.2
Operation Range		
ODU	Cooling (°F / °C)	-13~122 / -25~50
	Heating (°F / °C)	-13~75 / -25~24
IDU	Cooling (°F / °C)	60~90 / 16~32
	Heating (°F / °C)	32~86 / 0~30

Piping

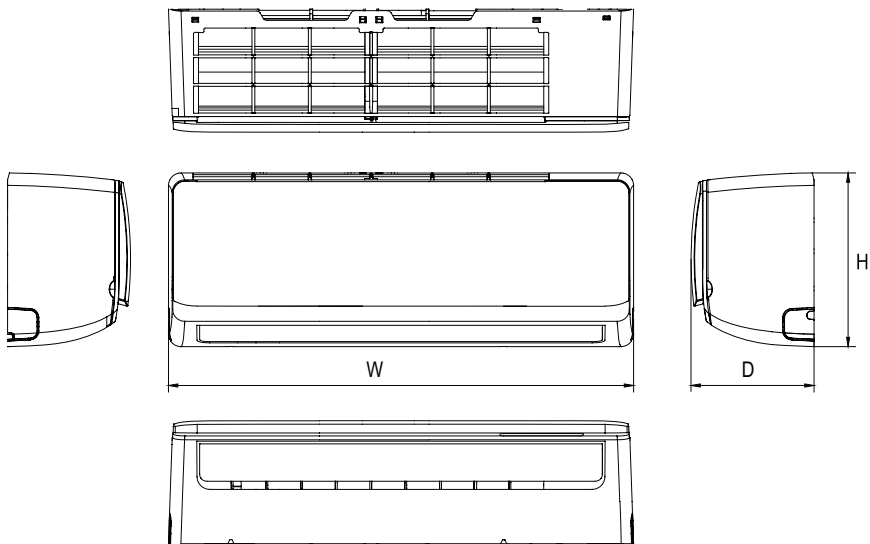
Refrigerant precharge (oz/kg)		47.97 / 1.36
Length for precharge (ft/m)		24.6 / 7.5
Liquid side (in/mm)		1/4" / Ø6.35
Gas side (in/mm)		1/2" / Ø12.7
Additional Charge of Refrigerant (oz/ft)/(g/m)		0.16 / 15
Max. refrigerant pipe length (ft/m)		98.4 / 30
Max. difference in height (ft/m)		65.6 / 20
System Data		
Refrigerant type		R454B
Design pressure (PSIG)		550 / 340
Dehumidification (pts/hr)		4.23
IDU	Fan motor model	ZKFP-30-8-3-10
	Fan motor qty (ea)	1
	Fan motor input (W)	36.0
	Fan speed (RPM)	1190 / 994 / 896
	Air flow, Turbo / H / M / L / Si (CFM)	635.7 / 500.3 / 359.0 / 294.3 / /
	Approx. Air Throw Range ft (m)	26.2 / 8
	Noise level, Turbo / H / M / L / Si (dB(A))	49 / 45 / 39 / 28 / 20
	Net weight (lbs / kg)	27.34 / 12.4
	Gross weight (lbs / kg)	36.60 / 16.6
ODU	Fan motor model	ZKFN-80-10-1L
	Fan motor qty (ea)	1
	Fan motor input (W)	161
	Fan motor speed, H / M / L (RPM)	780 / 700 / 550
	Air flow, Max (CFM)	1764.7
	Noise level, Max (dB(A))	58.0
	Net weight (lbs / kg)	92.59 / 42.0
	Gross weight (lbs / kg)	100.31 / 45.5
Accessories		
BMS-WT2-XXD (8733962707)		Wired Wall Thermostat
BMS-DRY-XXD (8733969263)		24V Interface
BMS-PK2-7X1 (8733951011)		25ft Lineset
Standard Parts		
8-733-969-088		Remote Control

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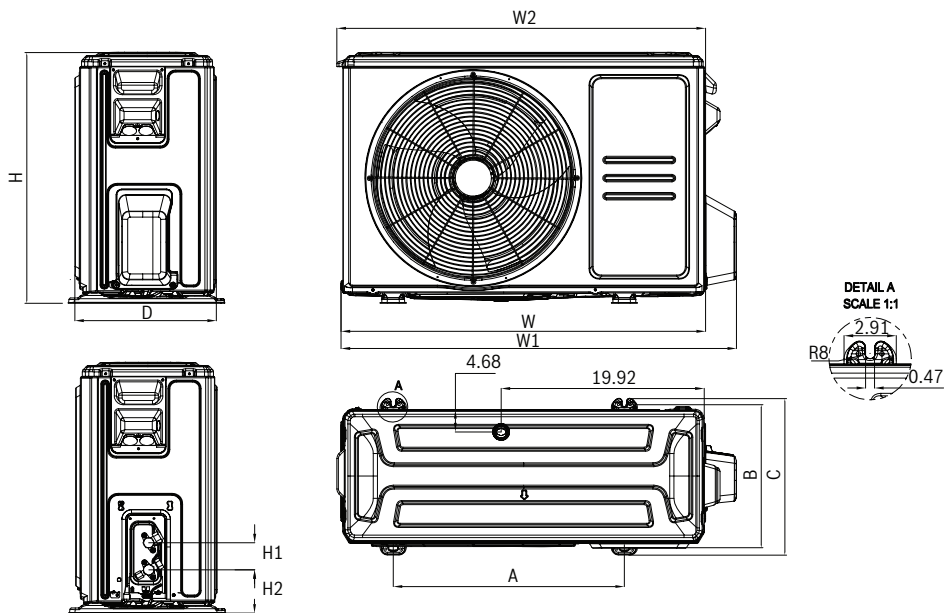


Indoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)
BMS500-AAU018-1AHWXD	38.0	9.8	12.56

Outdoor Unit Dimensions and Connections



Model	W (in)	D (in)	H (in)	W1 (in)	W2 (in)	H1 (in)	H2 (in)	A (in)	B (in)	C (in)
BMS500-AAS018-1CSXRD	35.04	13.46	26.50	37.6	35.24	2.36	4.25	26.1	13.7	14.96

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Cooling Performance - Indoor Dry Bulb Temperature at 635 CFM																					
ODU DB(°F)	IDU WB(°F)	60.8°F(16.0°C)					64.4°F(18.0°C)					67°F(19.4°C)					71.6°F(22.0°C)				
	IDU DB(°F)	73.4°F(23.0°C)	77.0°F(25.0°C)	80.0°F(26.7°C)	80.6°F(27.0°C)	84.2°F(29.0°C)	73.4°F(23.0°C)	77.0°F(25.0°C)	80.0°F(26.7°C)	80.6°F(27.0°C)	84.2°F(29.0°C)	73.4°F(23.0°C)	77.0°F(25.0°C)	80.0°F(26.7°C)	80.6°F(27.0°C)	84.2°F(29.0°C)	73.4°F(23.0°C)	77.0°F(25.0°C)	80.0°F(26.7°C)	80.6°F(27.0°C)	84.2°F(29.0°C)
-13°F (-25.0°C)	TC	15398	15391	15610	15675	16106	16065	16052	16042	16041	16027	16546	16525	16509	16507	16491	17453	17448	17412	17405	17369
	S/T	0.80	0.91	1.00	1.00	1.00	0.65	0.75	0.85	0.86	0.97	0.55	0.65	0.74	0.76	0.86	0.39	0.47	0.56	0.57	0.67
	PI	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67	0.67
-4°F (-20.0°C)	TC	16582	16575	16811	16881	17345	17301	17287	17276	17274	17260	17819	17796	17779	17777	17760	18795	18790	18751	18744	18705
	S/T	0.80	0.92	1.00	1.00	1.00	0.65	0.76	0.85	0.87	0.98	0.55	0.65	0.74	0.76	0.86	0.39	0.48	0.56	0.57	0.67
	PI	0.70	0.70	0.70	0.70	0.69	0.69	0.69	0.69	0.69	0.70	0.69	0.69	0.69	0.69	0.69	0.70	0.70	0.70	0.70	0.70
0°F (-17.8°C)	TC	17133	17122	17440	17513	17994	17867	17848	17834	17832	18011	18393	18366	18344	18341	18332	19382	19377	19376	19378	19391
	S/T	0.81	0.93	1.00	1.00	1.00	0.66	0.77	0.86	0.88	1.00	0.55	0.66	0.75	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.70	0.71	0.71	0.71	0.71	0.71
5°F (-15.0°C)	TC	17624	17685	18089	18172	18732	18450	18455	18456	18475	18751	19040	19045	19049	19050	19075	20157	20151	20162	20164	20173
	S/T	0.82	0.94	1.00	1.00	1.00	0.66	0.78	0.87	0.88	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.59	0.69
	PI	0.71	0.71	0.71	0.71	0.72	0.71	0.71	0.71	0.71	0.72	0.72	0.72	0.72	0.72	0.72	0.73	0.73	0.73	0.73	0.73
14°F (-10.0°C)	TC	18983	18974	19361	19452	20055	19926	19908	19895	19892	20079	20608	20582	20562	20558	20538	21913	21906	21864	21857	21810
	S/T	0.81	0.93	1.00	1.00	1.00	0.65	0.76	0.86	0.87	1.00	0.55	0.66	0.75	0.76	0.87	0.39	0.48	0.56	0.57	0.67
	PI	0.61	0.61	0.61	0.61	0.61	0.60	0.61	0.61	0.61	0.61	0.60	0.60	0.60	0.60	0.60	0.61	0.61	0.61	0.61	0.61
17°F (-8.3°C)	TC	19390	19453	19902	19999	20649	20355	20363	20367	20384	20671	21049	21057	21066	21067	21093	22387	22380	22377	22381	22400
	S/T	0.82	0.93	1.00	1.00	1.00	0.66	0.77	0.87	0.88	1.00	0.55	0.66	0.76	0.77	0.87	0.39	0.48	0.57	0.58	0.68
	PI	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57	0.57
23°F (-5.0°C)	TC	18855	18936	19426	19521	20165	19809	19815	19818	19844	20186	20493	20500	20505	20506	20539	21790	21783	21799	21802	21814
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.89	0.39	0.48	0.57	0.59	0.69
	PI	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.62	0.64	0.64	0.64	0.64	0.64
35°F (1.7°C)	TC	17956	17957	18062	18113	18697	18795	18795	18796	18796	18847	19384	19383	19384	19384	19386	20475	20470	20469	20469	20470
	S/T	0.78	0.88	0.97	0.99	1.00	0.63	0.73	0.82	0.83	0.93	0.54	0.63	0.72	0.73	0.83	0.39	0.46	0.54	0.56	0.65
	PI	0.67	0.67	0.67	0.67	0.70	0.71	0.71	0.71	0.71	0.71	0.74	0.74	0.74	0.74	0.74	0.80	0.80	0.80	0.80	0.80
41°F (5.0°C)	TC	18048	18048	18192	18318	18860	18887	18887	18888	18888	18954	19477	19475	19476	19477	19478	20569	20564	20563	20562	20564
	S/T	0.78	0.89	0.98	1.00	1.00	0.64	0.74	0.83	0.84	0.94	0.54	0.64	0.72	0.74	0.84	0.39	0.47	0.55	0.56	0.66
	PI	0.68	0.68	0.68	0.69	0.72	0.72	0.72	0.72	0.72	0.72	0.75	0.75	0.75	0.75	0.75	0.82	0.82	0.82	0.82	0.82
47°F (8.3°C)	TC	18139	18151	18394	18477	19023	18980	18979	18980	18980	19068	19569	19568	19569	19569	19569	20665	20660	20658	20657	20659
	S/T	0.79	0.90	1.00	1.00	1.00	0.64	0.75	0.84	0.85	0.96	0.54	0.65	0.73	0.75	0.85	0.39	0.47	0.55	0.57	0.67
	PI	0.69	0.69	0.70	0.71	0.74	0.73	0.73	0.73	0.73	0.74	0.77	0.77	0.77	0.77	0.77	0.84	0.84	0.84	0.84	0.84
50°F (10.0°C)	TC	17721	18209	18824	18933	19667	18722	18732	18862	18921	19686	19452	19478	19488	19489	19676	20793	20848	20902	20908	20934
	S/T	0.90	1.00	1.00	1.00	1.00	0.71	0.85	0.96	0.99	1.00	0.59	0.72	0.83	0.85	0.99	0.40	0.50	0.60	0.62	0.74
	PI	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79	0.79
59°F (15.0°C)	TC	17444	18056	18697	18811	19576	18457	18482	18692	18831	19596	19218	19238	19244	19272	19612	20637	20678	20719	20724	20741
	S/T	0.92	1.00	1.00	1.00	1.00	0.73	0.86	1.00	1.00	1.00	0.60	0.73	0.85	0.86	1.00	0.40	0.51	0.61	0.63	0.76
	PI	0.84	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.85	0.86	0.85	0.85	0.85	0.85
68°F (20.0°C)	TC	17161	17893	18557	18675	19467	18167	18208	18577	18695	19486	18953	18967	19001	19018	19502	20421	20459	20491	20495	20505
	S/T	0.95	1.00	1.00	1.00	1.00	0.74	0.89	1.00	1.00	1.00	0.61	0.75	0.87	0.89	1.00	0.40	0.51	0.63	0.65	0.78
	PI	0.90	0.90	0.91	0.91	0.91	0.90	0.90	0.91	0.91	0.91	0.91	0.91	0.91	0.91	0.91	0.92	0.92	0.92	0.92	0.92
77°F (25.0°C)	TC	16865	17677	18369	18491	19317	17879	17936	18389	18512	19337	18689	18710	18757	18778	19353	20185	20246	20293	20298	20316
	S/T	0.97	1.00	1.00	1.00	1.00	0.75	0.90	1.00	1.00	1.00	0.61	0.76	0.88	0.90	1.00	0.40	0.52	0.63	0.65	0.79
	PI	0.96	0.96	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.97	0.98	0.98	0.98	0.98	0.98
82°F (27.8°C)	TC	16796	17653	18365	18490	19337	17798	17867	18381	18511	19357	18626	18650	18708	18731	19374	20139	20215	20267	20273	20295
	S/T	0.99	1.00	1.00	1.00	1.00	0.76	0.91	1.00	1.00	1.00	0.62	0.77	0.89	0.92	1.00	0.40	0.52	0.64	0.66	0.80
	PI	1.06	1.07	1.07	1.07	1.08	1.07	1.07	1.07	1.07	1.08	1.08	1.07	1.07	1.07	1.08	1.09	1.09	1.09	1.09	1.08
86°F (30.0°C)	TC	16917	17762	18488	18617	19484	17841	17923	18509	18639	19505	18686	18708	18776	18803	19521	20230	20308	20356	20361	20380
	S/T	1.00	1.00	1.00	1.00	1.00	0.77	0.93	1.00	1.00	1.00	0.62	0.78	0.91	0.93	1.00	0.40	0.52	0.64	0.66	0.81
	PI	1.23	1.24	1.24	1.24	1.25	1.24	1.24	1.24	1.25	1.25	1.25	1.25	1.25	1.25	1.25	1.26	1.26	1.26	1.26	1.26
95°F (35.0°C)	TC	17065	17157	17712	17817	18521	18137	18147	18153	18174	18544	18915	18919	18930	18932	18964	20382	20374	20378	20384	20411
	S/T	0.83	0.95	1.00	1.00	1.00	0.66	0.78	0.88	0.89	1.00	0.56	0.67	0.76	0.78	0.88	0.39	0.48	0.57	0.58	0.68
	PI	1.55	1.55	1.58	1.58	1.61	1.60	1.60	1.60	1.60	1.61	1.63	1.63	1.63	1.63	1.63	1.70	1.70	1.70	1.70	1.70
104°F (40.0°C)	TC	14707	14842	15400	15493	16111	15634	15642	15670	15687	16131	16301	16307	16316	16317	16363	17570	17563	17571	17577	17596
	S/T	0.85	0.98	1.00	1.00	1.00	0.68	0.80	0.90	0.92	1.00	0.56	0.68	0.78	0.80	0.91	0.39	0.48	0.58	0.59	0.70
	PI	1.43	1.44	1.47	1.47	1.50	1.48	1.48	1.48	1.48	1.50	1.5									

Climate 5000 Series - Gen 4

18K Btu Single Zone Regular System - Wall Mounted IDU



Heating Performance - Indoor Dry Bulb Temperature at 635 CFM										
Outdoor DB(°F)	TCITotal Heating Capacity(BTU/hr)					PIPower Input(kW)				
	Indoor Conditions(DB °F)					Indoor Conditions(DB °F)				
	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)	60.8°F (16.0°C)	68°F (20.0°C)	70°F (21.1°C)	71.6°F (22.0°C)	75.2°F (24.0°C)
-13°F(-25.0°C)	9220	9036	8991	8955	8874	1.49	1.50	1.50	1.51	1.51
-4°F(-20.0°C)	11204	10980	10925	10882	10784	1.54	1.55	1.56	1.56	1.57
0°F(-17.8°C)	12330	12084	12023	11975	11867	1.57	1.58	1.58	1.59	1.59
5°F(-15.0°C)	13738	13463	13396	13342	13222	1.60	1.61	1.62	1.62	1.63
14°F(-10.0°C)	14918	14619	14546	14488	14358	1.74	1.76	1.76	1.76	1.77
17°F(-8.3°C)	15311	15005	14930	14870	14736	1.79	1.80	1.81	1.81	1.82
22°F(-5.6°C)	16127	15804	15725	15662	15521	1.87	1.88	1.89	1.89	1.90
27°F(-2.8°C)	16943	16604	16521	16455	16307	1.95	1.96	1.97	1.97	1.98
32°F(-0.0°C)	17759	17403	17316	17247	17092	2.03	2.04	2.05	2.05	2.06
35°F(1.7°C)	18248	17883	17794	17723	17563	2.08	2.09	2.10	2.10	2.11
37°F(2.8°C)	18791	18416	18324	18250	18086	2.04	2.06	2.07	2.07	2.08
42°F(5.6°C)	20150	19747	19648	19569	19393	1.97	1.98	1.99	1.99	2.00
44.6°F(7.0°C)	20693	20279	20178	20097	19916	1.94	1.95	1.95	1.96	1.97
47°F(8.3°C)	21508	21078	20973	20889	20701	1.89	1.90	1.91	1.91	1.92
52°F(11.1°C)	20949	20530	20427	20346	20163	1.67	1.69	1.69	1.70	1.70
57°F(13.9°C)	20390	19982	19882	19803	19625	1.46	1.47	1.48	1.48	1.49
59°F(15.0°C)	20166	19763	19664	19586	19409	1.38	1.39	1.39	1.39	1.40
62°F(16.7°C)	19831	19434	19337	19260	19087	1.25	1.26	1.26	1.26	1.27
64.4°F(18.0°C)	19563	19171	19076	18999	18828	1.15	1.16	1.16	1.16	1.16
69.8°F(21.0°C)	18959	18580	18487	18413	18247	0.92	0.92	0.92	0.93	0.93
75.2°F(24.0°C)	18355	17988	17898	17827	17666	0.69	0.69	0.69	0.69	0.70

Bolded values were tested. All other values are simulated and for reference only.